MATH MA'AM ACADEMY

Where Every Subject Clicks.

DR. HOMI BHABHA BALVAIDNYANIK COMPETITION • STANDARD VI

COMBINED REVISION NOTES

UNIT 2 — COMPOSITION OF SUBSTANCES | UNIT 3 — SEPARATION TECHNIQUES

PART A — COMPOSITION OF SUBSTANCES

1. MATTER

Matter is anything that has mass and occupies space. Matter can be classified into pure substances and mixtures.

2. PURE SUBSTANCES

A pure substance has a fixed composition and characteristic properties. The two major categories are elements and compounds.

Elements

• An element is a pure substance that cannot be broken down into two or more simpler substances by ordinary physical or chemical methods.

• Elements are classified as metals, non-metals and metalloids.

• Metals: generally lustrous, malleable, ductile and good conductors; examples include iron, copper, aluminium, silver and gold.

• Non-metals: generally poor conductors and may be solids, liquids or gases; examples include hydrogen, nitrogen, carbon and sulphur.

• Metalloids show properties intermediate between metals and non-metals; examples include silicon, boron, arsenic and germanium.

Classification of elements by atomic composition

Type	Meaning	Examples
Monoatomic	Elements whose particles exist as single atoms.	He, Ar, Ne, Kr, Xe
Diatomic	Elements whose molecules contain two atoms.	H■, N■, O■, Cl■
Polyatomic	Molecules containing more than two atoms.	O■, P■, S■

Compounds

• A compound is a pure substance formed when two or more elements chemically combine in a definite proportion by mass.

• Examples: water (H■O), sodium chloride (NaCl), iron sulphide (FeS), carbon dioxide (CO■).

• Compounds have properties different from those of their constituent elements.

• Constituent elements cannot be separated from a compound by simple physical methods; chemical methods are required.

• A pure compound has a fixed composition and definite melting and boiling points (at a given pressure).

3. MIXTURES

• A mixture contains two or more pure substances physically mixed in any proportion.

• The substances in a mixture are called components.

• Components generally retain their individual properties and can often be separated by physical methods.

• The composition of a mixture can vary.

Homogeneous mixture	Heterogeneous mixture
Uniform composition throughout; components are not visibly distinguishable.	Non-uniform composition; different parts may have different compositions.
Examples: salt solution, sugar solution, air, alcohol mixed with water.	Examples: oil and water, sand and salt, soil, muddy/turbid water.

Pure substance vs mixture

Pure substance	Mixture
Fixed composition.	Composition can vary.
One kind of substance: element or compound.	Two or more substances physically combined.
Definite characteristic melting/boiling point (for a pure material at fixed pressure).	Often melts or boils over a range.
A compound needs chemical methods for decomposition.	Components can often be separated physically.

PART B — SEPARATION TECHNIQUES

4. WHY DO WE SEPARATE MIXTURES?

Separation removes unwanted materials, obtains useful components, or obtains a pure substance. The method is selected according to differences in properties such as particle size, density, solubility, magnetism, boiling point or miscibility.

5. METHODS FOR SOLID–SOLID MIXTURES

Method	Principle / use	Example
Handpicking	Components differ in size, colour, shape or appearance and can be picked out manually.	Removing stones from rice or pulses.
Sieving	Particles of different sizes pass through or remain on a sieve.	Separating bran from flour; sand from gravel.
Magnetic separation	A magnet attracts magnetic material from non-magnetic material.	Iron filings from sulphur or sand.
Threshing	Separates grains from harvested stalks, usually by beating or mechanical action.	Separating wheat grains from stalks.
Winnowing	Air separates lighter chaff from heavier grains.	Separating husk/chaff from grain.

6. METHODS FOR SOLID–LIQUID MIXTURES

Method	Principle / use	Example
Sedimentation	Heavier insoluble particles settle at the bottom when the mixture is left undisturbed.	Mud settling in water.
Decantation	Carefully pouring off the clear liquid without disturbing settled sediment.	Pouring clear water off settled sand.
Filtration	A porous filter allows liquid through but retains insoluble solid particles.	Separating sand from water; filter paper.
Centrifugation	Rapid spinning separates components of different densities; denser particles move outward/settle faster.	Separating cream from milk; laboratory separation.
Evaporation	Liquid changes into vapour, leaving dissolved solid behind.	Obtaining salt from seawater.
Crystallisation	A dissolved substance forms pure crystals from a suitable solution, often by cooling or controlled evaporation.	Obtaining crystals of copper sulphate or salt.

7. METHODS FOR LIQUID–LIQUID MIXTURES

Method	Principle / use	Example
Separating funnel	Separates immiscible liquids that form distinct layers due to different densities.	Oil and water.
Distillation	Separates miscible liquids using differences in boiling points; vapour is cooled and condensed.	Separating suitable mixtures of alcohol and water; obtaining pure water from salt solution.

8. IMPORTANT DISTINCTIONS

Pair	Difference
Sedimentation vs decantation	Sedimentation is settling of heavier particles; decantation is pouring off the upper liquid.
Filtration vs evaporation	Filtration removes insoluble solids; evaporation recovers dissolved solid by vaporising the solvent.
Evaporation vs crystallisation	Evaporation may leave a residue; crystallisation is controlled formation of purer crystals.
Separating funnel vs distillation	Separating funnel is for immiscible liquids; distillation is used for miscible liquids with different boiling points.
Threshing vs winnowing	Threshing loosens/separates grains from stalks; winnowing uses air to separate lighter chaff from heavier grains.

9. CHOOSING THE CORRECT METHOD — QUICK GUIDE

Mixture / property	Suitable method
Different-sized solid particles	Sieving
Visible unwanted pieces	Handpicking
Magnetic + non-magnetic solids	Magnetic separation
Grain + stalk	Threshing
Grain + light chaff	Winnowing
Insoluble solid + liquid; solid settles	Sedimentation then decantation
Insoluble solid + liquid; fine particles	Filtration
Dissolved salt + water	Evaporation / crystallisation
Two immiscible liquids	Separating funnel
Two miscible liquids with different boiling points	Distillation

10. EXAM-FOCUSED CONCEPT CHECKS

• Salt dissolved in water forms a homogeneous mixture; sand mixed with water forms a heterogeneous mixture.

• Air is a homogeneous mixture of gases, not a compound.

• Water is a compound because hydrogen and oxygen are chemically combined in a fixed ratio.

• Iron filings can be removed from a mixture using a magnet; ordinary sulphur is non-magnetic.

• Evaporation of seawater leaves dissolved salts behind; in large-scale salt production, solar heat is commonly used.

• A separating funnel works only when the liquids are immiscible and form separate layers.

- Distillation involves evaporation followed by condensation; the condensed liquid is called the distillate.
- Filtration cannot separate dissolved salt from water because dissolved salt particles pass through ordinary filter paper.

Revision strategy: For every separation question, identify the physical property that differs first. Then select the technique that uses that difference.